

Q1.

The curve, C , has equation $y = \frac{x}{\cosh x}$

- (a) Show that the x -coordinates of any stationary points of C satisfy the equation $\tanh x = \frac{1}{x}$

$$\tanh x = \frac{1}{x}$$

(3)

- (b) (i) Sketch the graphs of $y = \tanh x$ and $y = \frac{1}{x}$ on the axes below.

$$y = \frac{x}{\cosh x}$$

(2)

- (ii) Hence determine the number of stationary points of the curve C .

(1)

- (c) Show that $\frac{d^2y}{dx^2} + y = 0$ at each of the stationary points of the curve C .

(4)

(Total 10 marks)

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